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Assessment – 2

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Task 2

Does normalization improve model quality on this dataset? Why or why not?

Pre-processing included testing normalization where a Z-score normalization was conducted on the data set. Since it avoids the situation where the model predicts based on the large value of any feature, this normalization most of the time enhances the quality of the model. Z-score normalization, however, only slightly improved this dataset of knowledge workers. However, normalization is still a common method that for some model and dataset properties selected may bring gains (Codecademy, 2022).

With normalization, are you able to get the model working with GradientDescentOptimizer? Z-score normalization doesn't seem to be able to get SGD working. Maybe a different type of normalization would?

The Stochastic Gradient Descent (SGD) optimizer was affected by Z-score normalization when it was unable to work effectively. This is attributed to the fact that Z-scoring only partly decreases the variance of measurements in possibly hard datasets for SGD to tackle. Other modifications that could enhance the result with SGD in this dataset are try using other forms of optimization or else will try a new kind of normalization with a dataset like min-max normalization.

Task 4

Pass at a model that uses all the features. Do you have any improvements?

Using all the numerical and categorical attributes, a primitive model was built first. In cross-validation as a comparison with models trained with only numeric's or categoricals, superior accuracy, precision, recall and F1 was obtained using ensemble methodology: however the efficiency of such models is still yet to be ascertained. More optimizations can be made by fine tuning factors like the number of samples passed through each epoch or the speed which the network learns, commonly known as the learning rate or, the architecture of the network including the number of layers or neurons in the layer.

Choice of the models

- DNN Regressor
- Linear Regressor

The Hyperparameters, Adam optimizer with learning rate of 0.005 was used. For feature representation, Categorical Features used categorical_column_with_vocabulary_list to apply

feature columns. Finally, all categorical and numerical features fed into the final DNN. For the same purpose of evaluating the non-linear models including DNN, a linear regression model was also implemented (Li et al., 2023, p. 1187-1199).

Model analysis and evaluation

Task 1 - Make your best model with numeric features

DNN Regressor

The initial loss rate of the first model, which included only numerical features, was rather high.

```
Call initializer instance with the dtype argument instead of passing it to the constructor
scores {'average_loss': 33751264.0, 'label/mean': 12949.43, 'loss': 532231500.0, 'prediction/mean': 12997.392, 'global_step': 1000}
WARNING:tensorflow:It seems that global step (tf.train.get_global_step) has not been increased. Current value (could be stable): 1106 vs pre
WARNING:tensorflow:It seems that global step (tf.train.get_global_step) has not been increased. Current value (could be stable): 1335 vs pre
scores {'average_loss': 27047986.0, 'label/mean': 12949.43, 'loss': 426525920.0, 'prediction/mean': 13009.029, 'global_step': 2000}
scores {'average_loss': 25129492.0, 'label/mean': 12949.43, 'loss': 396272770.0, 'prediction/mean': 13179.6875, 'global_step': 3000}
scores {'average_loss': 24405802.0, 'label/mean': 12949.43, 'loss': 384860700.0, 'prediction/mean': 13144.363, 'global_step': 4000}
scores {'average_loss': 23970986.0, 'label/mean': 12949.43, 'loss': 378004000.0, 'prediction/mean': 13077.541, 'global_step': 5000}
scores {'average_loss': 23623530.0, 'label/mean': 12949.43, 'loss': 372524900.0, 'prediction/mean': 13045.171, 'global_step': 6000}
scores {'average_loss': 23334180.0, 'label/mean': 12949.43, 'loss': 367962080.0, 'prediction/mean': 13003.627, 'global_step': 7000}
scores {'average_loss': 23057118.0, 'label/mean': 12949.43, 'loss': 363593020.0, 'prediction/mean': 12924.939, 'global_step': 8000}
scores {'average_loss': 22783236.0, 'label/mean': 12949.43, 'loss': 359274100.0, 'prediction/mean': 13044.417, 'global_step': 9000}
scores {'average_loss': 22595512.0, 'label/mean': 12949.43, 'loss': 356313820.0, 'prediction/mean': 12978.373, 'global_step': 10000}
```

Linear Regressor

Comparing results with DNN models and the baseline model with only linear numerical features, the first model has a significantly higher loss.

```
Evaluation scores: {'average_loss': 32382606.0, 'label/mean': 12949.43, 'loss': 510648800.0, 'prediction/mean': 13138.32, 'global_step': 2000}
Evaluation scores: {'average_loss': 27915120.0, 'label/mean': 12949.43, 'loss': 440199970.0, 'prediction/mean': 13114.801, 'global_step': 4000}
Evaluation scores: {'average_loss': 26260642.0, 'label/mean': 12949.43, 'loss': 414110140.0, 'prediction/mean': 13074.002, 'global_step': 6000}
Evaluation scores: {'average_loss': 25430426.0, 'label/mean': 12949.43, 'loss': 401018240.0, 'prediction/mean': 13149.098, 'global_step': 8000}
Evaluation scores: {'average_loss': 24925366.0, 'label/mean': 12949.43, 'loss': 393053860.0, 'prediction/mean': 13139.32, 'global_step': 10000}
Evaluation scores: {'average_loss': 24576556.0, 'label/mean': 12949.43, 'loss': 387553400.0, 'prediction/mean': 13051.2, 'global_step': 12000}
Evaluation scores: {'average_loss': 24289822.0, 'label/mean': 12949.43, 'loss': 383031800.0, 'prediction/mean': 13132.808, 'global_step': 14000}
Evaluation scores: {'average_loss': 24058516.0, 'label/mean': 12949.43, 'loss': 379384300.0, 'prediction/mean': 13157.483, 'global_step': 16000}
Evaluation scores: {'average_loss': 23856326.0, 'label/mean': 12949.43, 'loss': 376195900.0, 'prediction/mean': 13136.622, 'global_step': 18000}
Evaluation scores: {'average_loss': 23677698.0, 'label/mean': 12949.43, 'loss': 373379070.0, 'prediction/mean': 13095.369, 'global_step': 20000}
```

Task 3 - Make your best model using only categorical features

DNN Regressor

It was again founded that, the extent of loss of categorical feature-only model was significantly higher than that of the numeric feature-only model.

```
model_feature_columns [IndicatorColumn(categorical_column=VocabularyListCategoricalColumn(key='fuel-type', vocabulary_list=('gas', 'diesel')
scores {'average_loss': 216470880.0, 'label/mean': 12949.43, 'loss': 3413579300.0, 'prediction/mean': 648.0737, 'global_step': 2000}
scores {'average_loss': 201589890.0, 'label/mean': 12949.43, 'loss': 3178917600.0, 'prediction/mean': 1276.1346, 'global_step': 4000}
scores {'average_loss': 187934140.0, 'label/mean': 12949.43, 'loss': 2963577000.0, 'prediction/mean': 1883.6028, 'global_step': 6000}
scores {'average_loss': 175421150.0, 'label/mean': 12949.43, 'loss': 2766256600.0, 'prediction/mean': 2470.7607, 'global_step': 8000}
scores {'average_loss': 163968770.0, 'label/mean': 12949.43, 'loss': 2585661200.0, 'prediction/mean': 3037.9983, 'global_step': 10000}
scores {'average_loss': 153518060.0, 'label/mean': 12949.43, 'loss': 2420862000.0, 'prediction/mean': 3584.8047, 'global_step': 12000}
scores {'average_loss': 144007100.0, 'label/mean': 12949.43, 'loss': 2270881300.0, 'prediction/mean': 4110.8364, 'global_step': 14000}
scores {'average_loss': 135369730.0, 'label/mean': 12949.43, 'loss': 2134676400.0, 'prediction/mean': 4615.9614, 'global_step': 16000}
scores {'average_loss': 127540590.0, 'label/mean': 12949.43, 'loss': 2011217000.0, 'prediction/mean': 5100.1675, 'global_step': 18000}
scores {'average_loss': 120459256.0, 'label/mean': 12949.43, 'loss': 1899549800.0, 'prediction/mean': 5563.3496, 'global_step': 20000}
```

Linear Regressor

Considering only the categorical features, the average loss was very high as the linear regression model performed poorly as compared to the model that use all the features without transforming them into dummy variables.

```
WARNING:tensorflow:Using temporary folder as model directory: /tmp/tmp5kwk1q9p
Evaluation scores: {'average_loss': 32587954.0, 'label/mean': 12949.43, 'loss': 513886940.0, 'prediction/mean': 13090.74, 'global_step': 2000}
Evaluation scores: {'average_loss': 28064324.0, 'label/mean': 12949.43, 'loss': 442552800.0, 'prediction/mean': 13102.133, 'global_step': 4000}
Evaluation scores: {'average_loss': 26355548.0, 'label/mean': 12949.43, 'loss': 415606720.0, 'prediction/mean': 13176.706, 'global_step': 6000}
Evaluation scores: {'average_loss': 25506840.0, 'label/mean': 12949.43, 'loss': 402223260.0, 'prediction/mean': 13231.398, 'global_step': 8000}
Evaluation scores: {'average_loss': 24984162.0, 'label/mean': 12949.43, 'loss': 393981000.0, 'prediction/mean': 13186.588, 'global_step': 10000}
Evaluation scores: {'average_loss': 24618944.0, 'label/mean': 12949.43, 'loss': 388221800.0, 'prediction/mean': 13135.278, 'global_step': 12000}
Evaluation scores: {'average_loss': 24336648.0, 'label/mean': 12949.43, 'loss': 383770240.0, 'prediction/mean': 13118.839, 'global_step': 14000}
Evaluation scores: {'average_loss': 24105008.0, 'label/mean': 12949.43, 'loss': 380117440.0, 'prediction/mean': 13189.764, 'global_step': 16000}
Evaluation scores: {'average_loss': 23899690.0, 'label/mean': 12949.43, 'loss': 376879740.0, 'prediction/mean': 13163.026, 'global_step': 18000}
Evaluation scores: {'average_loss': 23718422.0, 'label/mean': 12949.43, 'loss': 374021280.0, 'prediction/mean': 13151.898, 'global_step': 20000}
```

Task 4 - Using all the features, make the best model

DNN Regressor

Thus, the numerical and categorical features improved the model's predictive performance, as the model had the least loss per DNN model.

```
WARNING:tensorflow:Using temporary folder as model directory: /tmp/tmpc6bo41us
WARNING:tensorflow:Using temporary folder as model directory: /tmp/tmpsg1ygffj
scores: {'average_loss': 222886740.0, 'label/mean': 12949.43, 'loss': 222101760.0, 'prediction/mean': 307.19592, 'global_step': 1000}
WARNING:tensorflow:Using temporary folder as model directory: /tmp/tmp3ok1lnye
scores: {'average_loss': 223304600.0, 'label/mean': 12949.43, 'loss': 222518900.0, 'prediction/mean': 291.22556, 'global_step': 1000}
WARNING:tensorflow:Using temporary folder as model directory: /tmp/tmpyki9m74c
scores: {'average_loss': 221564690.0, 'label/mean': 12949.43, 'loss': 220784560.0, 'prediction/mean': 347.85767, 'global_step': 1000}
WARNING:tensorflow:Using temporary folder as model directory: /tmp/tmpu5uxv6qz
scores: {'average_loss': 222667230.0, 'label/mean': 12949.43, 'loss': 221882700.0, 'prediction/mean': 315.31244, 'global_step': 1000}
WARNING:tensorflow:Using temporary folder as model directory: /tmp/tmpd8u0x3ys
scores: {'average_loss': 221579730.0, 'label/mean': 12949.43, 'loss': 220799260.0, 'prediction/mean': 347.45016, 'global_step': 1000}
WARNING:tensorflow:Using temporary folder as model directory: /tmp/tmp3y6_vhto
scores: {'average_loss': 221550930.0, 'label/mean': 12949.43, 'loss': 220771150.0, 'prediction/mean': 347.23932, 'global_step': 1000}
WARNING:tensorflow:Using temporary folder as model directory: /tmp/tmp9pn6v7n6
scores: {'average_loss': 223217230.0, 'label/mean': 12949.43, 'loss': 222433040.0, 'prediction/mean': 292.59918, 'global_step': 1000}
WARNING:tensorflow:Using temporary folder as model directory: /tmp/tmp0z7x4qpe
scores: {'average_loss': 222332670.0, 'label/mean': 12949.43, 'loss': 221550320.0, 'prediction/mean': 324.57123, 'global_step': 1000}
WARNING:tensorflow:Using temporary folder as model directory: /tmp/tmpk8775dah
scores: {'average_loss': 222269740.0, 'label/mean': 12949.43, 'loss': 221486880.0, 'prediction/mean': 326.1779, 'global_step': 1000}
scores: {'average_loss': 223543360.0, 'label/mean': 12949.43, 'loss': 222756030.0, 'prediction/mean': 287.945, 'global_step': 1000}
```

Linear Regressor

The improvement of the DNN model was greater than the linear regression model of all the features (Maulud & Abdulazeez, 2020, p. 140-147).

```
Evaluation scores: {'average_loss': 32030300.0, 'label/mean': 12949.43, 'loss': 505093200.0, 'prediction/mean': 13043.908, 'global_step': 2000}
Evaluation scores: {'average_loss': 27773510.0, 'label/mean': 12949.43, 'loss': 437966880.0, 'prediction/mean': 13120.549, 'global_step': 4000}
Evaluation scores: {'average_loss': 26182890.0, 'label/mean': 12949.43, 'loss': 412884030.0, 'prediction/mean': 13194.961, 'global_step': 6000}
Evaluation scores: {'average_loss': 25381404.0, 'label/mean': 12949.43, 'loss': 400245220.0, 'prediction/mean': 13141.433, 'global_step': 8000}
Evaluation scores: {'average_loss': 24891334.0, 'label/mean': 12949.43, 'loss': 392517200.0, 'prediction/mean': 13139.946, 'global_step': 10000}
Evaluation scores: {'average_loss': 24549564.0, 'label/mean': 12949.43, 'loss': 387127740.0, 'prediction/mean': 13070.546, 'global_step': 12000}
Evaluation scores: {'average_loss': 24272556.0, 'label/mean': 12949.43, 'loss': 382759550.0, 'prediction/mean': 13115.872, 'global_step': 14000}
Evaluation scores: {'average_loss': 24060304.0, 'label/mean': 12949.43, 'loss': 379412480.0, 'prediction/mean': 13007.142, 'global_step': 16000}
Evaluation scores: {'average_loss': 23847132.0, 'label/mean': 12949.43, 'loss': 376050900.0, 'prediction/mean': 13114.274, 'global_step': 18000}
Evaluation scores: {'average_loss': 23674758.0, 'label/mean': 12949.43, 'loss': 373332700.0, 'prediction/mean': 13055.308, 'global_step': 20000}
```

Overall findings

The DNN model that included all features gave the best results and hence the lowest average loss among all models used. This comparison between DNN and Linear regression highlights some of the only, usually general and numeric, data which can be processed by linear regression but not the DNN model.

References

Codecademy. (2022). Normalization. https://www.codecademy.com/article/normalization

Li, Z., Zhang, M., Xu, J., Yao, Y., Cao, C., Chen, T., Ma, X., & Lü, J. (2023). Lightweight approaches to DNN regression error reduction: An uncertainty alignment perspective. 2023 IEEE/ACM 45th International Conference on Software Engineering (ICSE), 1187-1199. https://doi.org/10.1109/icse48619.2023.00106

Maulud, D., & Abdulazeez, A. M. (2020). undefined. Journal of Applied Science and Technology Trends, 1(2), 140-147. https://doi.org/10.38094/jastt1457